

DATA301/DATA471 - Assignment IV

Luca Panettieri

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1 Exercise 1 - Estimators for the Exponential rate lambda

$X_1, \dots, X_n$ are independent and identically distributed as $\text{Exponential}(\lambda)$, where λ is greater than 0 and equals the reciprocal of the mean. I compare three estimators: the maximum likelihood estimator, the inter-quartile-range (IQR) analogy estimator $\hat{\lambda}_{IQR} = \ln(3)/(Q_3 - Q_1)$, and a bootstrap bias-corrected IQR estimator.

```
lambda_ml <- function(x) 1 / mean(x)

lambda_iqr <- function(x) {
  q <- quantile(x, c(0.25, 0.75), type = 7, names = FALSE)
  log(3) / (q[2] - q[1])
}
```

1.1 [10%] Read and summarise Wicklin (2017); which type of order statistics would you use and why, and which is the default in your R/Python?

Wicklin (2017) reviews nine definitions of the sample quantile: three discrete (which round to order statistics) and six continuous (which interpolate linearly between them). They differ mainly for small samples (n of 10 or fewer) and in the tails, and become indistinguishable by about $n = 100$. Defaults vary by software (SAS uses type 2; R uses type 7). Wicklin does not recommend a single type, but points to Hyndman and Fan (1996) for anyone who wants an unbiased or distribution-free choice. Hyndman and Fan recommend type 8, which is approximately median-unbiased for any distribution.

I use type 7, which is R's default (NumPy's "linear" method is the same). For continuous Exponential data with no ties the difference between the types is of order $1/n$ and cannot affect the comparison of estimators below, so the default is fine.

1.2 [20%] Derive and implement a first-order bootstrap bias-corrected estimator based on IQR ($B = 200$)

The IQR estimator is a nonlinear function of two quantiles, so it is biased in finite samples. Write θ_{hat} for the IQR estimate on the data. The bootstrap (Efron, 1979) resamples the data with replacement, recomputes the estimator to get $\theta_{\text{star}_1}, \dots, \theta_{\text{star}_B}$, and estimates the bias as their mean minus θ_{hat} . The first-order bias-corrected estimator subtracts that estimated bias:

$$\hat{\lambda}_{BC} = 2\hat{\theta} - \frac{1}{B} \sum_{b=1}^B \hat{\theta}_b^*, \quad B = 200.$$

```
lambda_iqr_bc <- function(x, B = 200) {
  theta_hat <- lambda_iqr(x)
  n <- length(x)
  theta_star <- replicate(B, lambda_iqr(x[sample.int(n, n, replace = TRUE)]))
  2 * theta_hat - mean(theta_star)
}
```

1.3 [20%] Make a Monte Carlo experiment comparing the three estimators (MSE as the sole metric); report results and justify the number of replications

Over λ in $\{0.1, 1, 10\}$ and n in $\{5, 30, 100\}$, with R replications per cell, I draw a fresh sample each replication and record the mean squared error (the sole metric), $\widehat{MSE} = \frac{1}{R} \sum_{r=1}^R (\hat{\lambda}_r - \lambda)^2$, together with its Monte Carlo standard error.

```

lambdas <- c(0.1, 1, 10); ns <- c(5, 30, 100)
R_MC <- 300

run_cell <- function(lambda, n, R = R_MC) {
  est <- matrix(NA_real_, R, 3, dimnames = list(NULL, c("ML", "IQR", "IQR_BC")))
  for (r in seq_len(R)) {
    x <- rexp(n, rate = lambda)
    est[r, ] <- c(lambda_ml(x), lambda_iqr(x), lambda_iqr_bc(x))
  }
  sq <- (est - lambda)^2
  data.frame(lambda, n, estimator = colnames(est),
             mse = colMeans(sq), mc_se = apply(sq, 2, sd) / sqrt(R),
             row.names = NULL)
}

results <- do.call(rbind, Map(run_cell,
                             expand.grid(lambda = lambdas, n = ns)$lambda,
                             expand.grid(lambda = lambdas, n = ns)$n))

mse_wide <- reshape(results[, c("lambda", "n", "estimator", "mse")],
                    idvar = c("lambda", "n"), timevar = "estimator", direction = "wide")
names(mse_wide) <- sub("^mse\\.", "", names(mse_wide))
knitr::kable(mse_wide, digits = 4, caption = "Estimated MSE by estimator for each (lambda, n).")

```

Table 1: Estimated MSE by estimator for each (lambda, n).

	lambda	n	ML	IQR	IQR_BC
1	0.1	5	0.0042	0.0837	Inf
4	1.0	5	0.4271	15.0686	Inf
7	10.0	5	38.5341	5688.0193	Inf
10	0.1	30	0.0004	0.0012	0.0013
13	1.0	30	0.0343	0.1031	0.1103
16	10.0	30	3.9449	13.7103	14.9941
19	0.1	100	0.0001	0.0002	0.0003
22	1.0	100	0.0098	0.0250	0.0292
25	10.0	100	1.0670	2.4792	2.7711

```

ggplot(results[is.finite(results$mse), ],
       aes(factor(n), mse, colour = estimator, group = estimator)) +
  geom_line() + geom_point() +
  facet_wrap(~ lambda, scales = "free_y", labeller = label_both) +
  scale_y_log10() +
  labs(x = "Sample size n", y = "MSE (log scale)", colour = "Estimator") +
  theme_minimal()

```

Replications. Following Robert and Casella (2009, section 3.2), a Monte Carlo average converges at a rate of order $1/\sqrt{R}$, and its error is judged from the estimated standard error: the estimate plus or minus two standard errors gives an approximate 95 percent band by the Central Limit Theorem (section 4.2 makes the same assessment by monitoring the variability of the runs). Applying that criterion at $R = 300$, the reported `mc_se` stays under about 16 percent of its MSE for every cell with n of 30 or more, which is still far smaller than the gaps between the estimators (ML versus IQR differs by well over 100 percent), so $R = 300$ is enough

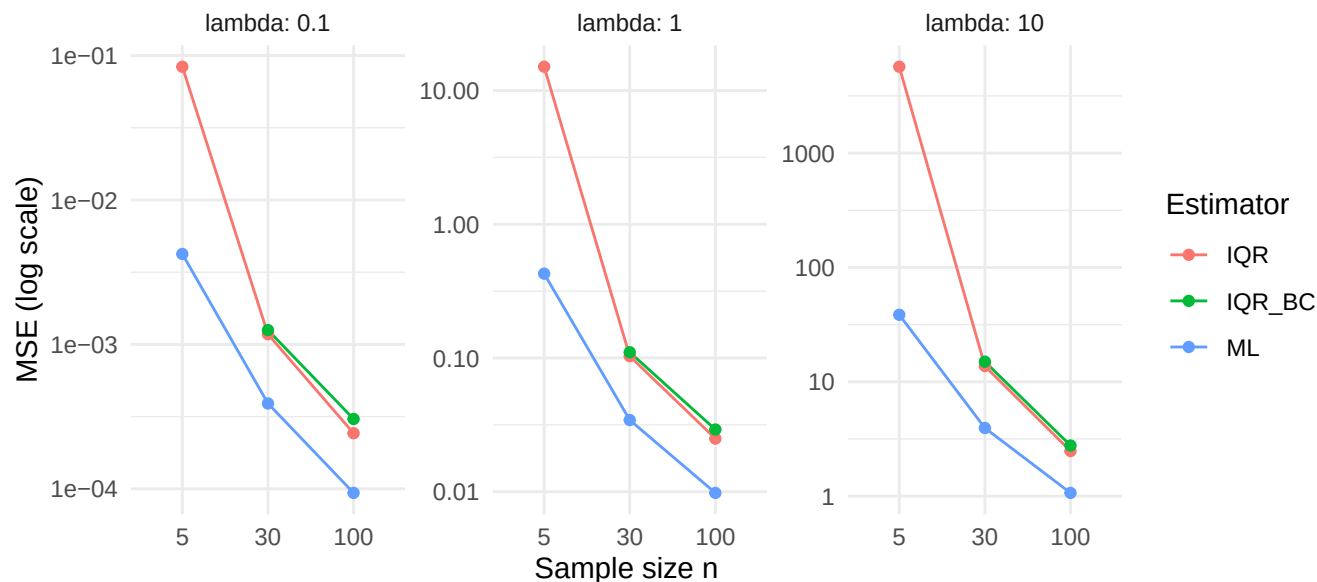


Figure 1: MSE against sample size, faceted by lambda (log scale). Non-finite points (IQR-BC at $n=5$) omitted.

to separate them. The small IQR versus IQR-BC gap is finer, but the three estimators are computed on the same sample in each replication (common random numbers, a variance-reduction technique), so their differences are estimated much more precisely than the individual standard errors suggest, and IQR-BC exceeds IQR in every finite cell. A larger R cannot help at $n = 5$, where the estimator is undefined rather than merely noisy.

Results. ML has the lowest MSE in every cell, which is expected because it is efficient when the model is correct. The IQR estimator is about 2.3 to 3.5 times worse than ML at n of 30 or more, since it uses only the quartiles. The bias correction slightly increases the MSE (it removes bias but adds more variance), and this holds in every finite cell. At $n = 5$ the IQR estimators fail: IQR is extremely variable, and IQR-BC is infinite, because bootstrap resamples of five values often give Q_3 equal to Q_1 (a zero denominator). The MSE scales like λ^2 across the columns and falls like $1/n$ down the rows, as expected.

2 Exercise 2 - Time-series analysis of ausbeer [50%]

This is approached as exploratory data analysis (per the course announcement): the plots below extract the information in the data, and no model is fitted. The ausbeer series is quarterly Australian beer production in megalitres; I use the span from 1992 onward.

```
beer <- window(ausbeer, start = 1992)
```

```
autoplot(beer) + labs(x = "Year", y = "Megalitres") + theme_minimal()
```

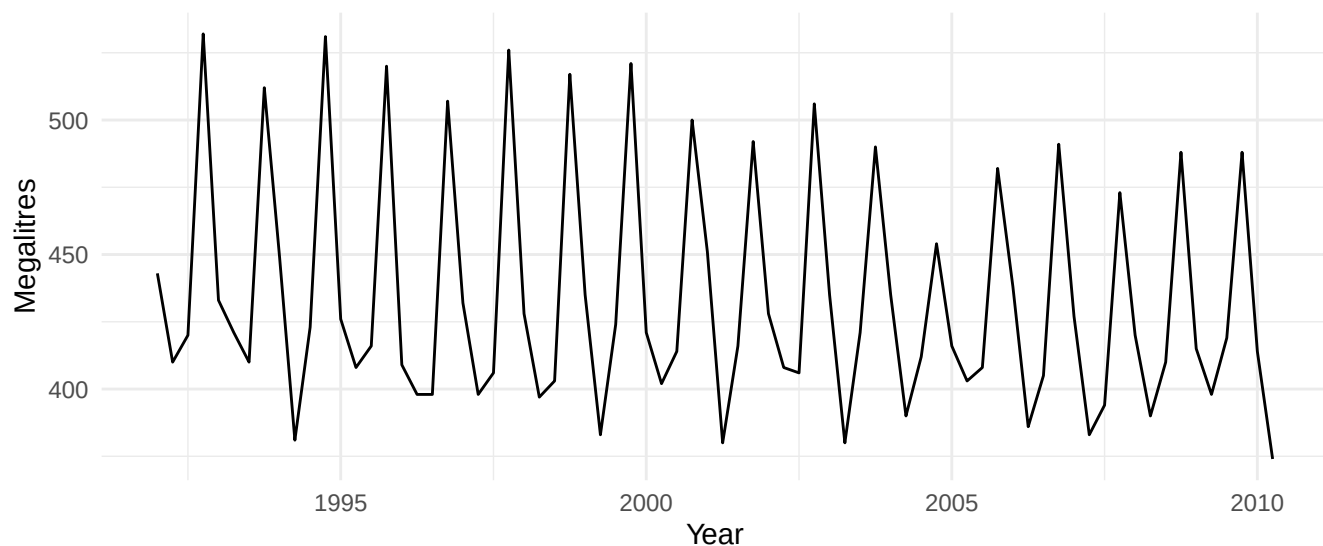


Figure 2: Quarterly Australian beer production, 1992 onward.

2.1 1. What is the frequency of the series? How can you determine this from the data and/or the time-series plot?

```
frequency(beer)
```

```
## [1] 4
```

The frequency is 4 (quarterly). This is reported by the series object, and the time plot above also repeats every four observations (one cycle per year).

2.2 2. Describe the overall behaviour of the series. Is there evidence of a long-term trend?

The series is broadly flat with a slight downward drift of about 1.3 megalitres per year. There is no strong long-term trend, just a gentle decline over the period.

2.3 3. Is there evidence of seasonality? If so, describe the seasonal pattern.

Yes. The seasonality is periodic (a fixed four-quarter cycle, not merely cyclic). The quarter means are Q1 429, Q2 394, Q3 411, Q4 502, so Q4 is the highest and Q2 the lowest, matching the southern-hemisphere Christmas and summer period.

2.4 4. Does the variability appear constant over time? Explain what you see in the plot and whether a transformation might be appropriate.

The seasonal amplitude looks stable across the whole window; the peaks and troughs do not fan out or shrink over time. Because the variability is roughly constant, a variance-stabilising transformation is not needed.

2.5 5. Are there any observations that you would regard as unusual? Identify them and explain why they attract attention.

The lowest point is 2010 Q2 (374 megalitres) and the highest is 1992 Q4 (532 megalitres). Both stand out as the extremes of the range, but they are consistent with the seasonal pattern (Q2 low, Q4 high) rather than being data errors.

2.6 6. Plot the series by season. What does this reveal that is less obvious in the ordinary time-series plot?

```
ggseasonplot(beer, year.labels = TRUE) + labs(y = "Megalitres") + theme_minimal()
```

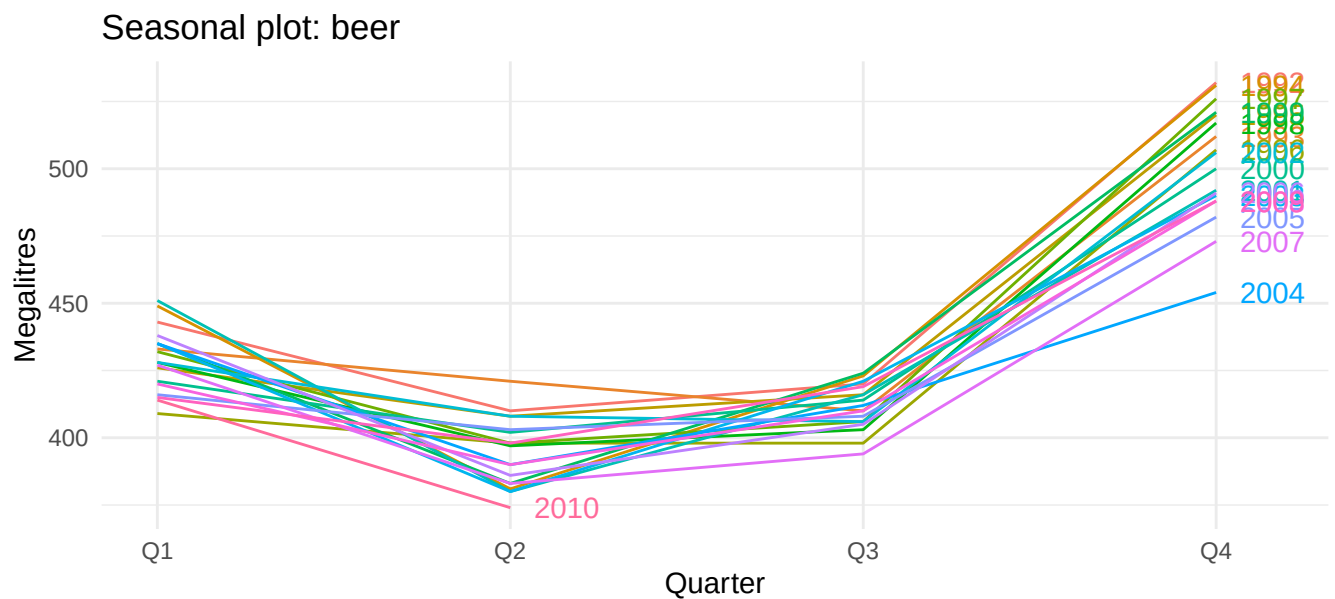


Figure 3: Seasonal plot.

Every year follows the same profile (a peak in Q4 and a trough in Q2) with roughly parallel lines. This makes it clear that the seasonal shape is stable and that years differ mainly in their overall level, which is hard to see in the ordinary plot.

2.7 7. Plot the seasonal subseries. Which quarters have the highest and lowest typical values? Is the seasonal pattern stable over time?

```
ggsubseriesplot(beer) + labs(y = "Megalitres") + theme_minimal()
```

Q4 has the highest typical value and Q2 the lowest (the blue quarter means). Each quarter drifts down slightly over time, but the ranking never changes, so the seasonal pattern is stable.

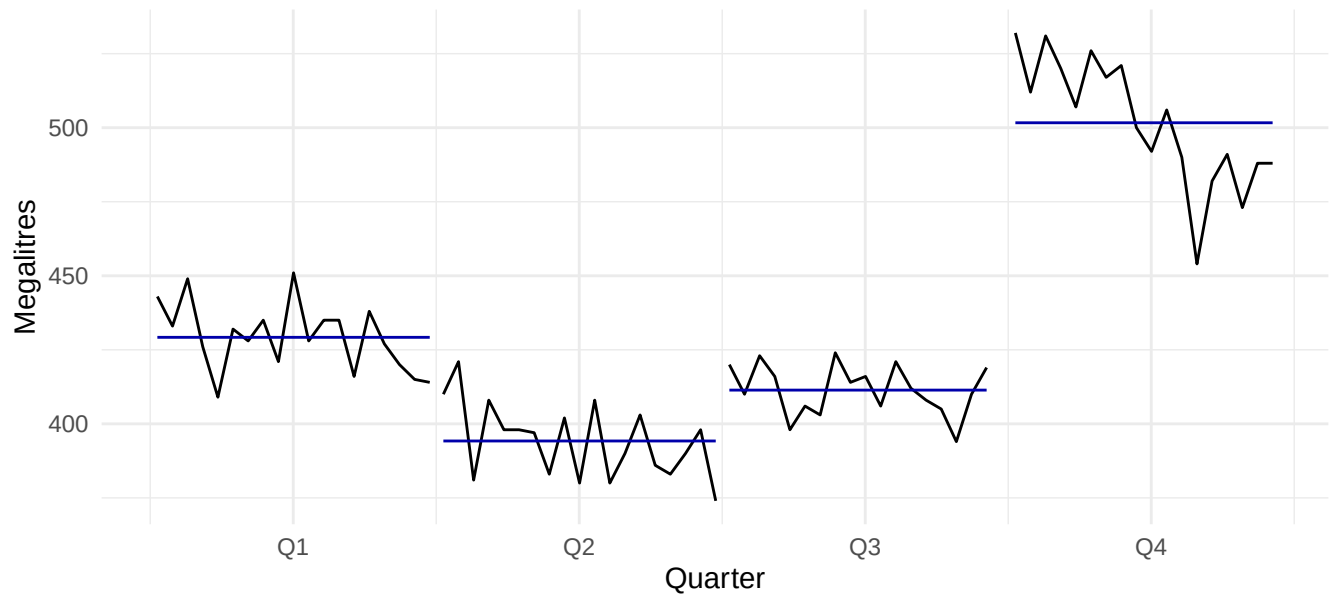


Figure 4: Seasonal subseries plot.

2.8 8. Would you detrend or transform the series before modelling it? Give a visual argument. Do not fit a model.

No. The variance is stable (so no transformation) and the trend is weak; the only substantial departure from stationarity is the seasonality, so if anything were done before modelling it would be a single seasonal difference. This is a visual argument only, with no model fitted.

3 References

- Efron, B. (1979). Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1), 1-26.
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- Robert, C. P., and Casella, G. (2009). *Introducing Monte Carlo Methods with R*. Springer.
- Wicklin, R. (2017). Sample quantiles: A comparison of 9 definitions. *SAS Blogs*. <https://blogs.sas.com/content/iml/2017/05/24/definitions-sample-quantiles.html>